BaitBreak - Scam Message Detection

Model Evaluation Report (Hindi / Hinglish / English)

1. Summary

A TF-IDF (word 1-2 grams + character 2-5 grams) text classifier was trained to label messages as **Legitimate (0)** or **Scam (1)**. Logistic Regression and Multinomial Naive Bayes were compared with 5-fold grouped cross-validation; **Logistic Regression** was selected (CV F1 0.8980). On the held-out test split it reached accuracy 96.8%, scam-class precision 86.3%, recall 95.8% and F1 90.8%. On a separate set of 72 unseen, hand-written Hindi/Hinglish/English messages the shipped model reached accuracy 90.3% and scam F1 90.4%.

Important caveat (read before relying on these numbers): the Hindi and Hinglish portions of the training data in this build are **template-based synthetic messages** written for this project, because the two Hugging Face datasets could not be downloaded in the build environment. English performance on the real UCI SMS data is a trustworthy estimate; Hindi/Hinglish figures are encouraging but come from small samples and should be re-measured after retraining with the real datasets (`fetch_external.py` then `build_dataset.py`) and with real BaitBreak user messages.

2. Dataset

Final cleaned dataset: **5,780** unique messages (label 0 = Legitimate, 1 = Scam). Hindi (Devanagari), Hinglish (Roman-script Hindi) and English text are all preserved.

Language	Legitimate	Scam	Total
English	4,610	724	5,334
Hinglish	117	110	227
Hindi	107	112	219

Source	Rows	Notes
UCI SMS Spam Collection	5,106	Real English SMS (ham/spam). Loaded from a GitHub mirror of the UCI file (5,574 raw rows).
Synthetic seed set	674	Hand-authored Indian-style scam/legit templates in Hindi, Hinglish and English (KYC, OTP, UPI, lottery, loan, job, customs, impersonation scams and genuine bank/OTP/delivery/personal messages).
Hinglish and English Financial Scam (HF)	0	Not included in this build (network blocked). Supported by build_dataset.py.
Indian Multilingual Scam (HF)	0	Not included in this build (network blocked). Supported by build_dataset.py.

Preparation steps

(1) Combine all sources; (2) drop empty / unlabelled rows; (3) clean text - Unicode NFC normalisation, remove zero-width characters, lower-case, mask URLs, e-mails, phone numbers and long digit strings with placeholder tokens, strip punctuation while keeping the whole Devanagari block; (4) remove duplicates on the cleaned text (and drop any text that appeared with both labels); (5) standardise labels to 0 = Legitimate, 1 = Scam; (6) tag each row with a language (Devanagari = Hindi, Roman Hindi vocabulary = Hinglish, otherwise English).

3. Method

Split. 80/20 grouped split: all messages generated from the same template are kept on the same side, so the test set measures generalisation to new wording instead of memorisation. Train = 4,626, test = 1,154.

Features. TF-IDF with sublinear term frequency: word 1-2 grams combined with character 2-5 grams (robust to Hinglish spelling variation and Devanagari).

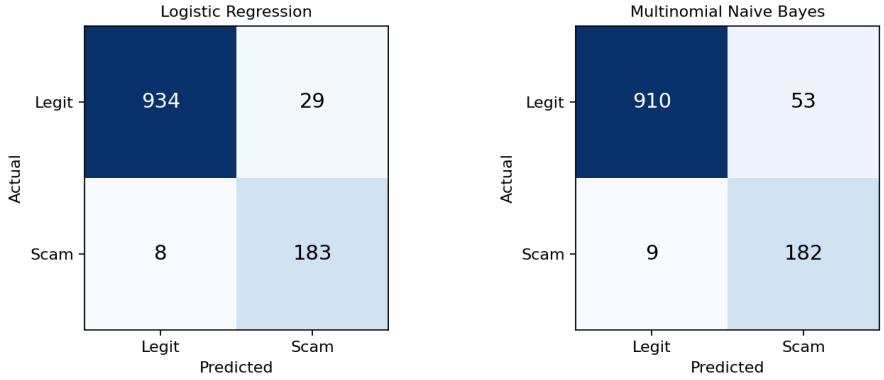
Models. Logistic Regression (balanced class weights, C tuned over {1, 5, 20}) and Multinomial Naive Bayes (alpha tuned over {0.05, 0.1, 0.5, 1.0}). Selection used 5-fold GroupKFold F1 of the scam class on the training split only; the test set played no role in selection. The selected configuration was then re-fitted on the full dataset for deployment (model.pkl, vectorizer.pkl).

4. Hold-out results

Positive class = Scam. Because most test messages are legitimate, accuracy alone is flattering; precision, recall and F1 of the scam class are the headline numbers.

Model	CV F1	Accuracy	Precision	Recall	F1
Logistic Regression (selected)	0.8980	0.9679	0.8632	0.9581	0.9082
Multinomial Naive Bayes	0.8777	0.9463	0.7745	0.9529	0.8545

Confusion matrix (test)	TN (legit ok)	FP (legit flagged)	FN (scam missed)	TP (scam caught)
Logistic Regression	934	29	8	183
Multinomial Naive Bayes	910	53	9	182



Selected model by slice (hold-out)

Slice	n	Accuracy	Precision	Recall	F1
language=english	1070	0.9748	0.8782	0.9448	0.9103
language=hindi	39	0.7949	0.7143	1.0000	0.8333
language=hinglish	45	0.9556	0.9286	1.0000	0.9630
source=uci_sms (real English SMS only)	1024	0.9902	0.9836	0.9375	0.9600

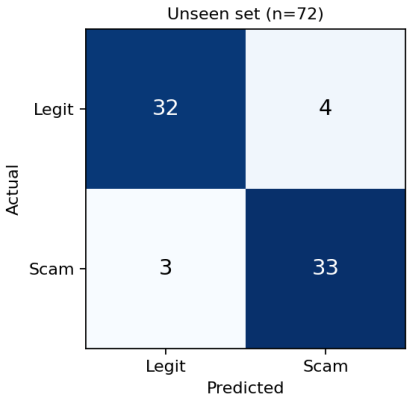
The Hindi and Hinglish slices contain only a few dozen messages, so their scores have wide error margins.

5. Unseen Hindi / Hinglish / English test

72 hand-written messages (12 scam + 12 legitimate per language) that are not in the training dataset and were not used for model selection. They include realistic legitimate bank/OTP alerts ("do not share OTP") that look superficially like scams. Note: these were written by the same team that wrote the seed set, so they are not fully independent of it; messages collected from real users will be a tougher test.

Subset	n	Accuracy	Precision	Recall	F1
Overall	72	0.9028	0.8919	0.9167	0.9041
English	24	0.8333	0.8333	0.8333	0.8333
Hinglish	24	0.9167	0.9167	0.9167	0.9167
Hindi	24	0.9583	0.9231	1.0000	0.9600

Confusion matrix (unseen)	TN	FP	FN	TP
Selected model	32	4	3	33



Misclassified messages

Lang	Actual	Predicted	Conf.	Message
english	scam	legit	0.81	Hi dear, I am stuck at the airport and my card is not working. Please transfer Rs 15000 to this UPI id, will pay back.
english	scam	legit	0.50	Your WhatsApp will be deactivated in 1 hour. Forward the 6-digit code you just received to verify.
english	legit	scam	0.55	Your OTP for Swiggy login is 482913. Do not share this code with anyone.
english	legit	scam	0.86	Rs 1,250 spent on your HDFC card ending 4521 at Big Bazaar on 02-Nov. Not you? Call the number on the back of your card.
hinglish	scam	legit	0.79	Aapka WhatsApp 1 ghante mein band ho jayega. Abhi aaye 6 digit code ko hume forward karo.
hinglish	legit	scam	0.90	Aapke HDFC card ending 4521 se Big Bazaar mein Rs 1,250 kharch hue 02-Nov ko. Aapne nahi kiya? Card ke peeche diye number par call karein.
hindi	legit	scam	0.95	[Hindi] Your HDFC card (4521) was charged Rs 1,250 at Big Bazaar on 02 Nov. Not you? Call the number on the back of the card.

(Hindi text is shown as an English gloss because the PDF font cannot render Devanagari; see the notebook for the original.)

Reading the errors

Two patterns dominate. (a) **Genuine bank debit alerts** that say "Not you? Call..." are flagged as scams because the wording overlaps with fake-debit scams; a genuine login-OTP message was also flagged, though only borderline (0.55). These are the most important things to fix with more real data, since false alarms on genuine bank/OTP messages hurt user trust. (b) **Novel scam stories** (stranded-friend money request,

WhatsApp code forwarding) were caught only with low confidence or missed, as similar examples are scarce in training.

6. Limitations and recommended next steps

- **Data provenance:** Hindi/Hinglish training data is synthetic in this build. Run `python fetch_external.py` and rebuild/retrain with the real Hugging Face datasets, then re-run the notebook and regenerate this report.
- **Add real examples:** collect and label real BaitBreak user messages (especially genuine bank alerts) and re-test; keep a frozen real-world test set.
- **Confidence is a model probability**, not a guarantee. Treat 0.5-0.7 as 'uncertain' in the UI (e.g. show a softer warning). The threshold is configurable via SCAM_THRESHOLD.
- **Scope:** classical bag-of-n-grams model; no URL reputation, sender, or conversation context. It should be one signal among several, not the only safeguard.
- **Drift:** scam wording changes quickly; plan periodic retraining and monitoring of false positives/negatives.